R22

Code No.: 22CSC21

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. / B.Tech (CSE/ IT/ AI&DS) V Sem (Main) Examination December 2024

Software Engineering

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A** at one place in the same order and **Part–B (Internal Choice)**

Part - A

(5Q X 2M = 10 Marks)

			M	CO	BT
1		What are the key outputs during an iteration in a project following an iterative development model?	(2)	1	1
2		What do you think happens when requirement validation uncovers an error? Who is involved in correcting the error?	(2)	2	1
3		List the problem based estimation guidelines.	(2)	3	1
4		What is the relationship between architecture and module-level design?	(2)	4	1
5		Is unit testing possible or even desirable in all circumstances? Provide examples to justify your answer.	(2)	5	2

Part - B

(5Q X 10M = 50 Marks)

			M	CO	BT
6	(a)	Umbrella activities occur throughout the software process. Do you think they are applied evenly across the process or are some concentrated in one or more framework activities.	(5)	1	2
	(b)	What are the Management Myths? Also state the reality that accompanies the Practitioner's myth.	(5)	1	2
		(OR)			
7	(a)	Illustrate the Waterfall model. Provide three examples of software projects that would be amenable to the waterfall model. Be specific.	(5)	1	2
	(b)	Describe the component based and formal methods for software development and list its advantages and disadvantages.	(5)	1	2
8	(a)	Most agile process models recommend face-to-face communication. Yet today, members of a software team and their customers may be geographically separated from one another. Do you think this implies that geographical separation is something to avoid? Can you think of ways to overcome this problem'.	(5)	2	3
	(b)	Why does an iterative process make it easier to manage change? Is every agile process being iterative? Explain your answers using Extreme Programming (XP) model.	(5)	2	2
		(OR)			
9	(a)	Discuss some of the problems that occur when requirements must be elicited from three or four different customers.	(5)	2	2
	(b)	Who should be involved in requirements review? Draw a process model showing how a requirements review might be organized.	(5)	2	3
10	(a)	Briefly explain the purpose of each of the sections in a software project plan.	(5)	3	2

	(b)	Cost estimates are inherently risky, irrespective of the estimation technique used. Suggest four ways in which the risk in a cost estimate can be reduced.	(5)	3	3
		(OR)			
11	(a)	What activities should be present in a project plan? Does your answer depend upon the software development life cycle used?	(5)	3	2
	(b)	Use the COCOMO II model to estimate the effort required to build software for a simple ATM that produces 12 screens, 10 reports, and will require approximately 80 software components. Assume average complexity and average developer/environment maturity. Use the application composition model with object points.	(5)	3	3
12	(a)	When describing a system, explain why you may have to design the system architecture before the requirements specification is complete.	(5)	4	2
	(b)	Consider a program containing many modules. If a global variable x must be used to share data between two modules A and B, how would you design the interfaces of these modules to minimize coupling? Explain your answer with one illustrative example.	(5)	4	3
		(OR)			
13	(a)	Provide the rules for evaluating the summary report of a code review.	(5)	4	2
	(b)	Describe the Principles and guidelines for Coding.	(5)	4	2
14	(a)	Describe the various White box testing methods for any one example system of your choice.	(5)	5	3
	(b)	Explain Software McCall's Quality Attributes.	(5)	5	2
		(OR)			
15	(a)	Explain why testing can only detect the presence of errors, not their absence.	(5)	5	2
	(b)	What are the benefits of involving users in release testing at an early stage in the testing process? Are there disadvantages in user involvement? Explain.	(5)	5	3